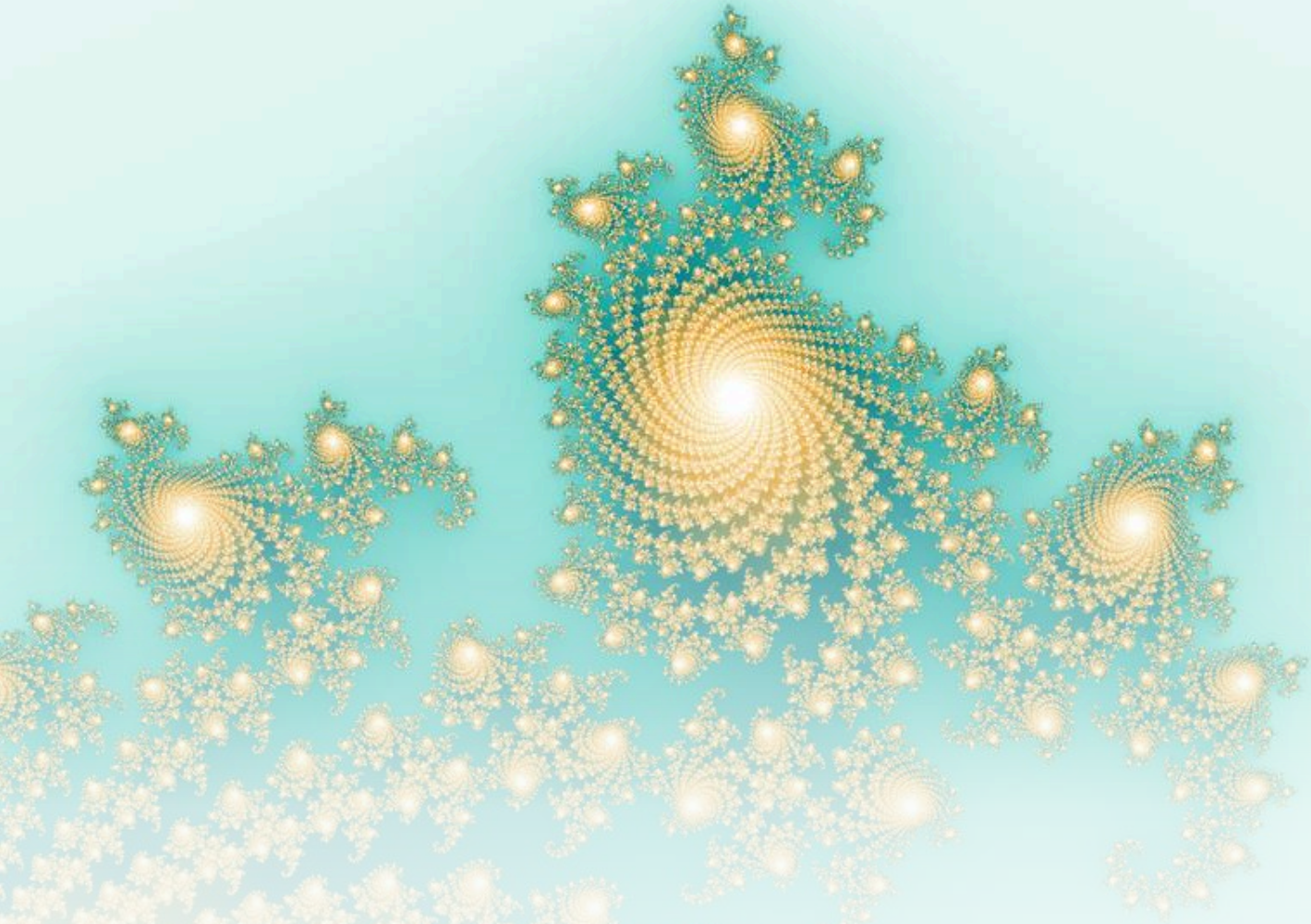


YEAR 8 STANDARD → YEAR 9 STANDARD

Summer Maths

Ready for Year 9

Six weeks. Twenty questions a week. Every Year 8 skill that Year 9 Standard builds on – mixed together so it sticks.

Week 1 | Week 2 | Week 3 | Week 4 | Week 5 | Week 6

NAME _____ CLASS _____

How it works

One page a week

Each week has **20 questions** and a **bonus**. Set aside about **40 minutes** – one sitting or a few short ones. Questions start easier and get a little harder through the page.

The same pattern every week

Question 1 is always integers, question 11 is always Pythagoras, and so on. If the same number keeps going wrong, that is the skill to revise before Year 9.

1	Integers	11	Pythagoras
2	Fractions & decimals	12	Area & perimeter
3	Percentages	13	Volume
4	Indices	14	Ratios & rates
5	Squares & roots	15	Number plane & lines
6	Simplify & substitute	16	Angles
7	Expand	17	Data
8	Factorise	18	Probability
9	Equations	19	Money in context
10	Formulas	20	Measurement in context

Why Julia? Each banner is part of the same *Julia set*: pick a number c, then repeat $z \to z^2 + c$ for every point. Points that stay small are coloured in. Every booklet in the series zooms into a different corner of the same set – this one looks straight down on its top spiral.

NSW syllabus (NESA Mathematics K–10, 2022). Revises MA4-INT-C-01, MA4-FRC-C-01, MA4-IND-C-01, MA4-ALG-C-01, MA4-EQU-C-01, MA4-LIN-C-01, MA4-PYT-C-01, MA4-LEN-C-01, MA4-ARE-C-01, MA4-VOL-C-01, MA4-RAT-C-01, MA4-ANG-C-01, MA4-GEO-C-01, MA4-DAT-C-02, MA4-PRO-C-01. ★ Year 9 preview (Stage 5 core): MA5-FIN-C-01, MA5-ALG-C-01, MA5-IND-C-01, MA5-EQU-C-01, MA5-LIN-C-01, MA5-LIN-C-02, MA5-TRG-C-01, MA5-DAT-C-01, MA5-PRO-C-01.

Rules of the game

- **No calculator.** Leave answers exact (e.g. 9π) unless asked to round.
- **Show working** in the space under each question.
- **Mark it yourself** using the answers at the back (pages 10–12). Each answer has a line of working to help you find a mistake.
- **Fix it.** Redo anything you got wrong a day or two later, without looking.
- ★ questions preview Year 9 (Stage 5) work. Have a go – they are the ones that will help most in Term 1.
- **Stuck?** The hints page (page 9) has the key facts and formulas.

My summer tracker

WEEK	DATE	SCORE	DONE	BONUS	REVISE (Q NUMBERS)
1 Week 1		/ 20	☐	☐	
2 Week 2		/ 20	☐	☐	
3 Week 3		/ 20	☐	☐	
4 Week 4		/ 20	☐	☐	
5 Week 5		/ 20	☐	☐	
6 Week 6		/ 20	☐	☐	

First up in Year 9 Standard: products and factors, Pythagoras' theorem, percentages and rates, trigonometry and indices. Every one of them starts from the skills in these six pages.

WEEK 1

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $-9 + 4 \times 3$

2 FRACTIONS & DECIMALS

Evaluate $\frac{5}{6} - \frac{1}{3}$

3 PERCENTAGES

Find 20% of \$45.

4 INDICES

Evaluate 3^4

5 SQUARES & ROOTS

Evaluate $\sqrt{64} - \sqrt[3]{27}$

6 SIMPLIFY & SUBSTITUTE

Simplify $6m + 2n - m + 5n$

7 EXPAND

Expand $5(x + 6)$

8 FACTORISE

Factorise $4x + 20$

9 EQUATIONS

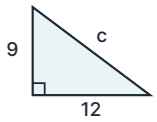
Solve $2x + 7 = 19$

10 FORMULAS

$A = lb$. Find A when $l = 12$ and $b = 7$.

11 PYTHAGORAS

Find c. (cm)



12 AREA & PERIMETER

Find the area of a triangle with base 14 cm and perpendicular height 5 cm.

13 VOLUME

Find the volume of a rectangular prism 8 cm by 5 cm by 3 cm.

14 RATIOS & RATES

Simplify the ratio $15 : 25$

15 NUMBER PLANE & LINES

Complete the table for $y = 3x + 1$.

x	-1	0	1	2
y				

16 ANGLES

Find a.



17 DATA

Find the mean of $5,\ 8,\ 6,\ 9,\ 7$.

18 PROBABILITY

A fair die is rolled. Find $P(\text{less than } 3)$.

19 MONEY IN CONTEXT ★

Jack works 12 hours at a surf shop and is paid \$22 an hour. How much does he earn?

20 MEASUREMENT IN CONTEXT

A garden bed 6 m by 2.5 m needs edging all the way around. How much edging is needed?

BONUS Find two numbers whose sum is 20 and whose difference is 6.

WEEK 2

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $-24 \div 6 - (-5)$

2 FRACTIONS & DECIMALS

Evaluate $2\frac{1}{3} \times \frac{3}{7}$

3 PERCENTAGES

Increase \$60 by 10%.

4 INDICES

Write 48 as a product of prime factors in index form.

5 SQUARES & ROOTS

Between which two consecutive whole numbers does $\sqrt{50}$ lie?

6 SIMPLIFY & SUBSTITUTE

If $a = 4$ and $b = -2$, find the value of $3a + 2b$.

7 EXPAND

Expand and simplify $2(x + 3) + 4x$

8 FACTORISE

Factorise $6y - 9$

9 EQUATIONS

Solve $\frac{x}{5} - 2 = 4$

10 FORMULAS

$P = 2(l + w)$. Find P when $l = 9$ and $w = 6$.

11 PYTHAGORAS

A right-angled triangle has shorter sides 5 cm and 12 cm. Find the length of the hypotenuse.

12 AREA & PERIMETER

Find the area of a circle with radius 3 cm, in terms of π.

13 VOLUME

A prism has a cross-section of area 24 cm² and length 10 cm. Find its volume.

14 RATIOS & RATES

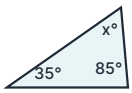
A car uses 8 L of fuel for every 100 km. How much fuel does it use for 350 km?

15 NUMBER PLANE & LINES

Does the point $(2,\ 9)$ lie on the line $y = 4x + 1$?

16 ANGLES

Find x.



17 DATA

Find the median of $12,\ 7,\ 9,\ 15,\ 10,\ 8$.

18 PROBABILITY

A bag holds 3 red, 5 green and 4 yellow marbles. Find $P(\text{not green})$.

19 MONEY IN CONTEXT

A surfboard costs \$440, including 10% GST. How much of the price is GST?

20 MEASUREMENT IN CONTEXT

A water tank holds 4.5 kL. How many litres is this?

BONUS The angles of a triangle are $x°$, $2x°$ and $3x°$. Find x.

WEEK 3

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $(-3)^2 + (-2)^3$

2 FRACTIONS & DECIMALS

Evaluate 0.8×0.05

3 PERCENTAGES

Write 18 out of 24 as a percentage.

4 INDICES

Evaluate $2^3 \times 5^2$

5 SQUARES & ROOTS

Evaluate $\sqrt{9} \times \sqrt{16}$

6 SIMPLIFY & SUBSTITUTE

Simplify $3x \times 5y$

7 EXPAND

Expand $-3(2x - 4)$

8 FACTORISE

Factorise $12a + 18b$

9 EQUATIONS

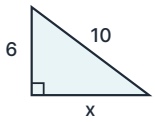
Solve $5x - 3 = 2x + 9$

10 FORMULAS

$v = u + at$. Find v when $u = 4$, $a = 3$ and $t = 6$.

11 PYTHAGORAS

Find x. (cm)



12 AREA & PERIMETER

Find the area of a trapezium with parallel sides 7 cm and 11 cm and perpendicular height 6 cm.

13 VOLUME

Find the volume of a cylinder with radius 3 cm and height 10 cm, in terms of π.

14 RATIOS & RATES

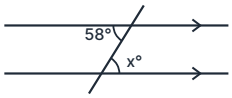
Share \$80 in the ratio $3 : 5$.

15 NUMBER PLANE & LINES ★

Find the gradient of the interval joining $(1,\ 3)$ and $(4,\ 9)$.

16 ANGLES

The lines are parallel. Find x.



17 DATA

Find the range and the mode of $4,\ 9,\ 7,\ 9,\ 2,\ 6$.

18 PROBABILITY

The probability that a spinner lands on red is 0.45. Find $P(\text{not red})$.

19 MONEY IN CONTEXT ★

Ava earns \$25 an hour. She works 8 hours at the normal rate and 2 hours at time-and-a-half. Find her pay.

20 MEASUREMENT IN CONTEXT

A pool is 10 m long, 5 m wide and 1.5 m deep. How many litres does it hold? ($1\ \text{m}^3 = 1000$ L)

BONUS

Show that the sum of two consecutive odd numbers is always a multiple of 4.

WEEK 4

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $-5 \times (-4) - 30 \div 6$

2 FRACTIONS & DECIMALS

Evaluate $\frac{3}{4} \div \frac{1}{8}$

3 PERCENTAGES

Decrease \$150 by 30%.

4 INDICES

Write as a single power: $4^6 \times 4^2$

5 SQUARES & ROOTS

Estimate $\sqrt{40}$ to the nearest whole number.

6 SIMPLIFY & SUBSTITUTE

If $x = -3$, find the value of $x^2 - 2x$.

7 EXPAND ★

Expand $(x + 2)(x + 5)$

8 FACTORISE

Factorise $8x^2 + 12x$

9 EQUATIONS

Solve $4(x - 3) = 20$

10 FORMULAS

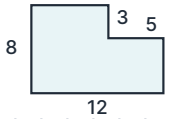
$F = 1.8C + 32$. Find F when $C = 25$.

11 PYTHAGORAS

Is a triangle with sides 8 cm, 15 cm and 17 cm right-angled? Explain.

12 AREA & PERIMETER

Find the area. (cm)



13 VOLUME

A cube has a volume of 64 cm³. How long is each edge?

14 RATIOS & RATES

A cyclist travels 45 km in 1.5 hours. Find the average speed.

15 NUMBER PLANE & LINES

Find the point where the lines $y = 2x$ and $y = x + 3$ meet.

16 ANGLES

Three angles of a quadrilateral are $100°$, $75°$ and $90°$. Find the fourth.

17 DATA ★

Find the interquartile range of $2,\ 4,\ 5,\ 7,\ 8,\ 10,\ 11,\ 13$.

18 PROBABILITY

A letter is chosen at random from the word SYDNEY. Find $P(\text{Y})$.

19 MONEY IN CONTEXT

A \$40 beach umbrella is reduced by 25%. Find the sale price.

20 MEASUREMENT IN CONTEXT

A circular trampoline mat has a diameter of 4 m. Find its area in terms of π.

BONUS

A rectangle's length is 3 cm more than its width, and its perimeter is 26 cm. Find its area.

WEEK 5

Summer Maths

Ready for Year 9

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $(-6 + 2) \times (-3) - 4$

2 FRACTIONS & DECIMALS

Evaluate $1\frac{3}{4} + 2\frac{5}{8}$

3 PERCENTAGES

An item bought for \$80 is sold for \$100. Find the profit as a percentage of the cost price.

4 INDICES

Evaluate $7^0 + 2^5$

5 SQUARES & ROOTS

Solve $x^2 = 49$

6 SIMPLIFY & SUBSTITUTE ★

Simplify $3x^2 \times 4x^3$

7 EXPAND

Expand and simplify $4(2x - 1) - 3(x + 2)$

8 FACTORISE

Factorise $15mn - 10m$

9 EQUATIONS

Solve $\dfrac{x + 4}{3} = 5$

10 FORMULAS

$A = lb$. Find b when $A = 54$ and $l = 9$.

11 PYTHAGORAS

A 5 m ladder leans against a wall with its foot 3 m from the wall. How high up the wall does it reach?

12 AREA & PERIMETER

Find the area of a kite with diagonals 10 cm and 7 cm.

13 VOLUME

A triangular prism has a triangle with base 6 cm and height 5 cm as its cross-section, and length 12 cm. Find its volume.

14 RATIOS & RATES

Which is the better buy: 500 g for \$4.50 or 750 g for \$6.00?

15 NUMBER PLANE & LINES ★

Find the gradient of the interval joining $(-2,\ 5)$ and $(2,\ -3)$.

16 ANGLES

Find x.



17 DATA

The mean of five numbers is 12. Four of them are 8, 15, 10 and 14. Find the fifth.

18 PROBABILITY ★

Two coins are tossed. Find $P(\text{two heads})$.

19 MONEY IN CONTEXT ★

Mia earns a salary of \$62 400 a year. How much is she paid each week? (1 year = 52 weeks)

20 MEASUREMENT IN CONTEXT

A 2 m by 3 m rug is placed in a room 5 m by 4 m. What area of floor is not covered?

BONUS Expand and simplify $(x + 3)^2 - x^2$

Date started _____

Score ____ / 20

Done ☐ Bonus ☐

Number · Indices & roots · Algebra · Equations · Measurement · Rates & graphs · Angles, data & chance · In context

★ Year 9 preview (Stage 5) · No calculator

1 INTEGERS

Evaluate $-48 \div (-8) + (-2) \times 7$

2 FRACTIONS & DECIMALS

Evaluate $2\frac{1}{2} \div 1\frac{1}{4}$

3 PERCENTAGES

After a 20% discount, a shirt costs \$36. What was the original price?

4 INDICES ★

Evaluate 2^{-3}

5 SQUARES & ROOTS

Evaluate $\sqrt{6^2 + 8^2}$

6 SIMPLIFY & SUBSTITUTE

Simplify $18a^2b \div 6ab$

7 EXPAND ★

Expand $(x - 4)(x + 3)$

8 FACTORISE

Factorise $-6x - 15$

9 EQUATIONS ★

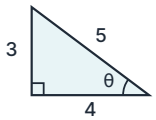
Solve $3(2x - 1) = 4x + 9$

10 FORMULAS

$A = \frac{1}{2}(a + b)h$. Find h when $A = 40$, $a = 6$ and $b = 10$.

11 PYTHAGORAS ★

For angle θ, name the opposite side and write $\tan\theta$ as a fraction.



12 AREA & PERIMETER

Find the perimeter of a semicircle with radius 6 cm, in terms of π.

13 VOLUME

A cylinder has a volume of 200π cm³ and a radius of 5 cm. Find its height.

14 RATIOS & RATES

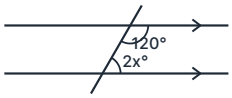
A map has a scale of $1 : 50\,000$. Two towns are 6 cm apart on the map. How far apart are they, in km?

15 NUMBER PLANE & LINES ★

Write the equation of the line with gradient 2 and y-intercept -3.

16 ANGLES

The lines are parallel. Find x.



17 DATA

For $3,\ 4,\ 4,\ 5,\ 34$, which is most affected by the outlier: the mean, median or mode?

18 PROBABILITY

A whole number from 1 to 30 is chosen at random. Find $P(\text{multiple of 4 or 6})$.

19 MONEY IN CONTEXT

Leo invests \$5000 at 4% p.a. simple interest. How much interest does he earn in 3 years?

20 MEASUREMENT IN CONTEXT

A cylindrical tank has radius 1 m and height 2 m. Find its capacity in litres, in terms of π. ($1\ \text{m}^3 = 1000$ L)

BONUS

A right-angled triangle has hypotenuse 5 and shorter sides x and $x + 1$. Find x.

Hints & formulas

Number

Order of operations: brackets, powers, × and ÷, then + and −

$\frac{a}{b} \div \frac{c}{d} = \frac{a}{b} \times \frac{d}{c}$

Increase by 10%: $\times 1.1$ · decrease by 30%: $\times 0.7$

% profit $= \dfrac{\text{profit}}{\text{cost price}} \times 100\%$

Price with GST is 110% of the price without GST

Indices & roots

$a^m \times a^n = a^{m+n}$ · $a^m \div a^n = a^{m-n}$ · $a^0 = 1$

$\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

$x^2 = 49 \Rightarrow x = 7$ or $x = -7$

★ $2^{-3} = \dfrac{1}{2^3} = \dfrac{1}{8}$

Algebra

Like terms: same letters, same powers

$a(b + c) = ab + ac$ · $-(x - 5) = -x + 5$

Factorise: take out the HCF, then check by expanding

★ $(a + b)(c + d) = ac + ad + bc + bd$

Equations & formulas

Do the same to both sides

Unknowns on both sides: collect them on one side first

Formulas: substitute, then solve

Check your answer by substituting it back in

Measurement

Pythagoras: $c^2 = a^2 + b^2$ (c = hypotenuse)

$A = \frac{1}{2}bh$ · $A = \frac{h}{2}(a + b)$ · kite $A = \frac{1}{2}xy$

Circle: $C = 2\pi r$, $A = \pi r^2$

Prism $V = Ah$ · cylinder $V = \pi r^2 h$

$1\ \text{m}^3 = 1000\ \text{L}$ · $1\ \text{kL} = 1000\ \text{L}$

★ $\sin\theta = \frac{O}{H}$, $\cos\theta = \frac{A}{H}$, $\tan\theta = \frac{O}{A}$

Rates & graphs

Speed = distance ÷ time

Share in a ratio: add the parts, find one part

A point is on a line if it fits the equation

★ Gradient $= \dfrac{\text{rise}}{\text{run}}$; $y = mx + c$: m = gradient, c = y-intercept

Angles, data & chance

Straight line $180°$ · point $360°$ · vertically opposite angles are equal

Parallel lines: corresponding, alternate equal; co-interior add to $180°$

Triangle $180°$ · quadrilateral $360°$

Median: middle value in order · $P(\text{not } E) = 1 - P(E)$

★ IQR $= Q_3 - Q_1$ (medians of the lower and upper halves)

In context

Simple interest $I = Prn$ (r as a decimal)

★ Time-and-a-half: $1.5\times$ the hourly rate

★ Weekly pay = yearly salary ÷ 52

Worded problems: write what you know, choose the operation, answer with units

Answers

1 Week 1 answers

1 3 multiply first: $-9+12$

2 $\frac{1}{2}$ $\frac{5}{6}-\frac{2}{6}=\frac{3}{6}$

3 **$9** $10\%=4.50$; double it

4 81 $3\times3\times3\times3$

5 5 $8-3$

6 $5m+7n$ $6m-m$ and $2n+5n$

7 $5x+30$ $5\times x+5\times6$

8 $4(x+5)$ HCF = 4

9 $x=6$ $2x=12$

10 $A=84$ 12×7

11 **15 cm** $c^2=81+144=225$

12 **35 cm²** $\frac{1}{2}\times14\times5$

13 **120 cm³** $8\times5\times3$

14 $3:5$ divide both parts by 5

15 $-2,\ 1,\ 4,\ 7$ e.g. $3(-1)+1=-2$

16 $a=35$ angles on a straight line: $180-145$

17 7 $35\div5$

18 $\frac{1}{3}$ $\{1,2\}$: $\frac{2}{6}$

19 **$264** 12×22

20 **17 m** $2(6+2.5)$

B **13 and 7** $13+7=20,\ 13-7=6$

2 Week 2 answers

1 1 $-4+5$

2 1 $\frac{7}{3}\times\frac{3}{7}$

3 **$66** $60+6$

4 $2^4\times3$ $48=16\times3$

5 **7 and 8** $49<50<64$

6 8 $12+(-4)$

7 $6x+6$ $2x+6+4x$

8 $3(2y-3)$ HCF = 3

9 $x=30$ $\frac{x}{5}=6$

10 $P=30$ 2×15

11 **13 cm** $25+144=169$

12 **9π cm²** $\pi\times3^2$

13 **240 cm³** $V=Ah=24\times10$

14 **28 L** 3.5×8

15 **Yes** $4(2)+1=9$ ✓

16 $x=60$ angle sum: $180-35-85$

17 9.5 7, 8, **9, 10**, 12, 15: $\frac{9+10}{2}$

18 $\frac{7}{12}$ 7 of the 12 are not green

19 **$40** $440 is 110%; 1% = 4, so 10% = 40

20 **4500 L** $1\text{ kL}=1000\text{ L}$

B $x=30$ $6x=180$; angles $30°, 60°, 90°$

Answers

3 Week 3 answers

1 1 $9 + (-8)$

2 0.04 $8 \times 5 = 40$; 3 decimal places: 0.040

3 75% $\frac{18}{24} = \frac{3}{4}$

4 200 8×25

5 12 3×4 (or $\sqrt{144}$)

6 $15xy$ $3 \times 5 = 15$

7 $-6x + 12$ $-3 \times 2x$ and $-3 \times (-4)$

8 $6(2a + 3b)$ HCF = 6

9 $x = 4$ $3x - 3 = 9$, $3x = 12$

10 $v = 22$ $4 + 18$

11 **8 cm** $x^2 = 100 - 36 = 64$

12 **54 cm²** $\frac{6}{2}(7 + 11) = 3 \times 18$

13 **90π cm³** $\pi \times 9 \times 10$

14 **$30 and $50** 8 parts: $10 each

15 2 $\frac{\text{rise}}{\text{run}} = \frac{9-3}{4-1} = \frac{6}{3}$

16 $x = 58$ alternate angles are equal

17 **range 7, mode 9** range $9 - 2$

18 0.55 $1 - 0.45$

19 **$275** $8 \times 25 + 2 \times 37.50 = 200 + 75$

20 **75 000 L** $10 \times 5 \times 1.5 = 75\ \text{m}^3$

B $(2n + 1) + (2n + 3) = 4n + 4 = 4(n + 1)$ a multiple of 4 for every whole number n

4 Week 4 answers

1 15 $20 - 5$

2 6 $\frac{3}{4} \times 8$

3 **$105** 30% of 150 = 45; $150 - 45$

4 4^8 add the indices

5 6 $6^2 = 36$ and $7^2 = 49$; 40 is closer to 36

6 15 $9 - (-6) = 9 + 6$

7 $x^2 + 7x + 10$ $x^2 + 5x + 2x + 10$

8 $4x(2x + 3)$ HCF = $4x$

9 $x = 8$ $x - 3 = 5$

10 $F = 77$ $45 + 32$

11 **Yes** $64 + 225 = 289 = 17^2$

12 **81 cm²** $12 \times 8 - 5 \times 3 = 96 - 15$

13 **4 cm** $4 \times 4 \times 4 = 64$

14 **30 km/h** $45 \div 1.5$

15 $(3,\ 6)$ $2x = x + 3$, so $x = 3$

16 95° $360 - 265$

17 6 $Q_1 = 4.5$, $Q_3 = 10.5$

18 $\frac{1}{3}$ 2 Y's out of 6 letters

19 **$30** 25% of 40 = 10

20 **4π m²** $r = 2$: $\pi \times 2^2$

B **40 cm²** width 5, length 8: $2(5 + 8) = 26$

Answers

5 Week 5 answers

1 8 $(-4)(-3) - 4 = 12 - 4$

2 $4\frac{3}{8}$ $\frac{14}{8} + \frac{21}{8} = \frac{35}{8}$

3 25% $\frac{20}{80} \times 100\%$

4 33 $1 + 32$

5 $x = 7$ **or** $x = -7$ $7^2 = 49$ and $(-7)^2 = 49$

6 $12x^5$ 3×4; add the indices

7 $5x - 10$ $8x - 4 - 3x - 6$

8 $5m(3n - 2)$ HCF = $5m$

9 $x = 11$ $x + 4 = 15$

10 $b = 6$ $54 = 9b$

11 **4 m** $h^2 = 25 - 9 = 16$

12 **35 cm²** $\frac{1}{2} \times 10 \times 7$

13 **180 cm³** $A = 15$; $V = 15 \times 12$

14 **750 g for $6.00** $0.90 vs $0.80 per 100 g

15 -2 $\frac{-3-5}{2-(-2)} = \frac{-8}{4}$

16 $x = 20$ vertically opposite: $3x + 10 = 70$

17 **13** total $5 \times 12 = 60$; $60 - 47$

18 $\frac{1}{4}$ HH, HT, TH, TT: 1 of 4

19 **$1200** $62\,400 \div 52$

20 **14 m²** $20 - 6$

B $6x + 9$ $x^2 + 6x + 9 - x^2$

6 Week 6 answers

1 -8 $6 + (-14)$

2 2 $\frac{5}{2} \times \frac{4}{5}$

3 **$45** $36 is 80%; 10% = 4.50

4 $\frac{1}{8}$ $\frac{1}{2^3}$

5 10 $\sqrt{36 + 64} = \sqrt{100}$

6 $3a$ $\frac{18}{6} = 3$; $\frac{a^2b}{ab} = a$

7 $x^2 - x - 12$ $x^2 + 3x - 4x - 12$

8 $-3(2x + 5)$ take out -3

9 $x = 6$ $6x - 3 = 4x + 9$, $2x = 12$

10 $h = 5$ $40 = 8h$

11 **opposite = 3;** $\tan\theta = \frac{3}{4}$ $\tan\theta = \frac{\text{opposite}}{\text{adjacent}}$

12 $(6\pi + 12)$ **cm** half of $2\pi \times 6$, plus the diameter 12

13 **8 cm** $200\pi = \pi \times 25 \times h$

14 **3 km** $6 \times 50\,000 = 300\,000$ cm

15 $y = 2x - 3$ $y = mx + c$

16 $x = 30$ co-interior: $2x + 120 = 180$

17 **the mean** mean 10; median 4; mode 4

18 $\frac{1}{3}$ 4, 6, 8, 12, 16, 18, 20, 24, 28, 30: $\frac{10}{30}$

19 **$600** $I = Prn = 5000 \times 0.04 \times 3$

20 2000π **L** $V = \pi \times 1^2 \times 2 = 2\pi$ m^3

B $x = 3$ $x^2 + (x + 1)^2 = 25$; sides 3, 4, 5